Negation Flips max to min: Fard & Pineau (2011), (20)→(21) — *Student Fill-In Version*

*Instructions: the structure, equations, and step labels are given. Replace each [**YOUR ANSWER**] with your own explanation. If you can fill every one correctly, you understand the proof.*

1 The Original Result

What is the evaluation MDP M', and how are A' and R' defined in terms of the original?
[YOUR ANSWER]

Theorem 1 (Fard & Pineau, Thm. 1)**.**

$$Q^{\Pi}_{M}(s,a) \;=\; -\,Q^{*}_{M'}(s,a).$$

In one line, what does the proof do to get from the Bellman equation on M' to the claim?
[YOUR ANSWER]

$$Q^{*}_{M'}(s,a) = \bar{R}'(s,a) + \gamma \sum_{s'} T(s,a,s') \max_{a' \in A'} Q^{*}_{M'}(s',a') \tag{19}$$

$$\Rightarrow\; -Q^{*}_{M'}(s,a) = -\bar{R}'(s,a) - \gamma \sum_{s'} T(s,a,s') \max_{a' \in A'} Q^{*}_{M'}(s',a') \tag{20}$$

$$\Rightarrow\; -Q^{*}_{M'}(s,a) = \bar{R}(s,a) + \gamma \sum_{s'} T(s,a,s') \min_{a' \in \Pi(s')} \big(-Q^{*}_{M'}(s',a')\big). \tag{21}$$

Between (20) and (21), which changes are purely definitional substitutions, and which single change is the real inference?
[YOUR ANSWER]

State the one substantive identity being used:

[YOUR ANSWER]

What hypothesis (from Definition 1) is implicit here but cannot be skipped in a formalization? Why is it needed?
[YOUR ANSWER]

2 The Formalization, Step by Step

How does the bare identity translate: what do the action set, the Q-value, and max / min *become? What does the primed* sup'/inf' *require as an argument, and why?*
[YOUR ANSWER]

$$-\operatorname*{sup'}_{a\in A} Q(a) \;=\; \operatorname*{inf'}_{a\in A}\bigl(-Q(a)\bigr). \tag{$\star$}$$

Overall strategy

To prove an equality of two reals, what do we prove instead, and how are the two directions related?
[YOUR ANSWER]

Reshape. *What does a reshape step do, using which identities, and does it change what is true?*
[YOUR ANSWER]

Close. *What kind of fact closes a branch? List the four defining facts about* max / min *used.*
[YOUR ANSWER]

Direction 1: $-\operatorname{sup'}_a Q \le \operatorname{inf'}_a(-Q)$

1. **Reduce to elements.** *Why does the RHS being a minimum let us reduce to a per-element statement?*

   [YOUR ANSWER]

$$-\operatorname*{sup'}_{a'} Q(a') \;\le\; -Q(a). \qquad (\forall\, a \in A)$$

2. **Fix an arbitrary** a. *Why does proving it for a generic* a *suffice for all elements? How is this different from a loop?*

   [YOUR ANSWER]

3. **Reshape.** *What is the shape of the goal, which identity applies, and what does the goal become?*

   [YOUR ANSWER]

$$Q(a) \;\le\; \operatorname*{sup'}_{a'} Q(a').$$

4. **Close.** *Which defining fact finishes this, and why does it match?*

   [YOUR ANSWER]

Direction 2: $\operatorname{inf'}_a(-Q) \le -\operatorname{sup'}_a Q$

How does this direction relate to Direction 1? **[YOUR ANSWER]**

1. **Reshape #1.** *What shape is the goal, which identity applies, and what gets exposed?*

   [YOUR ANSWER]

$$\operatorname*{sup'}_{a'} Q(a') \;\le\; -\operatorname*{inf'}_{a'}\bigl(-Q(a')\bigr).$$

2. **Reduce to elements.** *Why can we now reduce to a per-element statement?*

   [YOUR ANSWER]

$$Q(a) \;\le\; -\operatorname*{inf'}_{a'}\bigl(-Q(a')\bigr). \qquad (\forall\, a \in A)$$

3. **Fix an arbitrary a.**
4. **Reshape #2.** *Same identity as #1 — what does it expose this time?*

   [YOUR ANSWER]

$$\operatorname*{inf'}_{a'}\bigl(-Q(a')\bigr) \;\le\; -Q(a).$$

5. **Close.** *Which defining fact finishes this, and against which element?*

   [YOUR ANSWER]

With both directions done, what lets us conclude the equality ($\star$), and how does ($\star$) relate to the paper's (20)→(21) step?
[YOUR ANSWER]

Two remarks

- **"Fix an arbitrary a" is not a loop.** *Explain.*

  [YOUR ANSWER]

- **Non-emptiness is load-bearing.** *Explain what breaks without it, and how the mechanization treats it.*

  [YOUR ANSWER]

A Lean 4 Source — annotate each line yourself

For each tactic below, note: is it a Reshape *or a* Close*, and what is the goal before/after? (Full source intentionally given; the comments are removed so you supply them.)*

```
import Mathlib
open Finset

theorem neg_max_eq_min_neg
    {Action : Type*}
    (actions : Finset Action)
    (h_nonempty : actions.Nonempty)
    (Q : Action -> R) :
    - actions.sup' h_nonempty Q
      = actions.inf' h_nonempty (fun x => - Q x) := by
  apply le_antisymm
  . apply Finset.le_inf'
    intro a ha
    rw [neg_le_neg_iff]
    exact Finset.le_sup' Q ha
  . rw [le_neg]
    apply Finset.sup'_le
    intro a ha
    rw [le_neg]
    exact Finset.inf'_le (fun x => - Q x) ha
```

Questions to answer for the appendix:

- Why does `apply le_antisymm` produce exactly two goals? **[YOUR ANSWER]**
- What does `intro a ha` destructure, and where does the type of `a` come from? **[YOUR ANSWER]**
- Why is `h_nonempty` passed around even though it contributes no number to the answer? **[YOUR ANSWER]**
- In the last line, what are the two arguments to `Finset.inf'_le`, and what does the applied lambda reduce to? **[YOUR ANSWER]**